



Semester 1 Examinations 2011 / 2012

Exam Code(s)	1SD1, 1MF1, 1IT1
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Master in Information Technology
Module Code(s)	CT516 CT861
Module(s)	Computer Architecture and Operating Systems
Paper No.	I
External Examiner(s)	Prof. Michael O'Boyle
Internal Examiner(s)	Professor Gerard Lyons Dr. Jim Duggan *Dr. Michael Schukat

Instructions: Answer 4 questions. All questions carry equal marks.

Duration	2 hrs
No. of Pages	4
Discipline(s)	Information Technology
Requirements	None

Question 1

- a) Distinguish between *distributed, embedded, desktop and mobile computers*. Where can these types of computers be found in the areas of *medicine* or *everyday activities*? Provide two examples per computer type.
10 MARKS
- b) Characterise and contrast the *operating systems* and *hardware* you find in *distributed, embedded, desktop and mobile computers*.
10 MARKS
- c) What kind of *interrupts* and *interrupt sources* would you find in an
- embedded computer
 - desktop computer?
- 5 MARKS

Question 2

- a) What is the idea of the *stored program concept* as introduced by John von Neumann? How does this concept fit into the *von Neumann architecture*?
9 MARKS
- b) What are in your view the advantages and disadvantages of the *Harvard architecture* over the *von Neumann architecture*?
9 MARKS
- c) In recent years there has been a trend towards *multi-core microprocessors*. What are the challenges with this from an *operating system design* and *application programming* point of view?
7 MARKS

Question 3

- a) Assume you want to connect a *single dynamic RAM chip* with 512 Byte (i.e. 512 x 8 bit) capacity to a CPU. Discuss how many *control, address and data lines* the interconnecting bus would require?
9 MARKS
- b) Using a diagram show how an I/O module can connect a keyboard to CPU. What suitable *I/O technique* would you apply to read the keyboard (e.g. a key that has been pressed by a user)?
9 MARKS
- c) Discuss and contrast the various *external interfaces* you find in a modern PC.
7 MARKS

Question 4

- a) Even though interrupts as asynchronous events can occur any time, they are only registered / processed by a CPU during certain well defined processor states. Why is this necessary and how is this achieved by a CPU?
8 MARKS
- b) A compiler translates a high level program into a sequence of CPU-specific instructions. What kind of program fragments (written in “C”) are translated into the following CPU instruction types:
- Data transfer
 - Arithmetic
 - Logical
 - Conversion
 - Transfer of control
- Provide one example per instruction type to illustrate your answer.
10 MARKS
- c) Assume you design a CPU with *16-bit instructions* where each instruction has *2 operand references*. Draw up a *suitable instruction format* and calculate how many instructions and operand references you can distinguish.
7 MARKS

Question 5

- a) Distinguish between the terms *paging* and *demand paging*. Provide an example that shows the advantage of the latter over the first.
9 MARKS
- b) Modern page replacement policies include *first in-first out* and *least recently used*. Do you think a *last in-first out* policy makes sense? Provide an example in support of your answer.
9 MARKS
- c) Discuss the relationship between *page size* and *internal fragmentation*. Why is it not feasible to eliminate memory wastage due to internal fragmentation?
7 MARKS

Question 6

- a) Using an example outline the need for the *7 state process model* (e.g. the process model that incorporates 2 *suspend states*).
9 MARKS
- b) *Stack, heap, data* and *text* are memory areas of a process. Write a small program in C to show how and why these memory areas are needed.
9 MARKS
- c) Imagine a virus or other malicious code that can access and modify all fields of the *process control blocks* of all processes in a running computer system. What damage could it cause?
7 MARKS

Question 7

- a) Using an example show the need for *process priorities* in an operating system.
7 MARKS
- b) *Two concurrently executed processes* need access the same non-shareable resource. How can you make sure that only one process accesses the resource at a time? Use code fragments or diagrams to illustrate your answer.
10 MARKS
- c) Outline the benefits and problems of the following disk scheduling policies:
- FIFO
 - SSTF
 - SCAN
 - C-SCAN
- 8 MARKS

Question 8

- a) Discuss in detail, how the *write policy, block size, mapping function* and *replacement algorithm* influence the efficiency of a cache.
10 MARKS
- b) What is meant by term *locality of reference*? Discuss if and how a good programming style can take advantage of this concept.
8 MARKS
- c) Modern hard disks have a build-in *semiconductor cache* (e.g. a static RAM). Why in your opinion is it needed and how does it work?
7 MARKS